

GUANGCHANG, China

WMO: 588130

Lat: 26.8379N

Lon: 116.3189E

Elev: 142

StdP: 99.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.8	(c) 0.4	(d) -6.6	(e) 2.2	(f) 6.4	(g) -4.2	(h) 2.7	(i) 5.5	(j) 5.3	(k) 8.3	(l) 4.7	(m) 8.7	(n) 0.8	(o) 180	(p) 0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.6	(c) 36.2	(d) 26.1	(e) 35.2	(f) 26.0	(g) 34.2	(h) 25.8	(i) 27.3	(j) 32.8	(k) 27.0	(l) 32.4	(m) 26.6	(n) 31.8	(o) 2.4	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.9	29.0	25.7	21.3	28.7	25.4	20.9	28.6	87.7	33.0	86.0	32.5	84.4	32.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.1	(b) 4.3	(c) 3.7	(e) -3.3	(f) 37.9	(g) 1.5	(h) 0.9	(i) -4.3	(j) 38.6	(k) -5.2	(l) 39.1	(m) -6.1	(n) 39.6	(o) -7.1	(p) 40.2		
			DB													
			WB													
				-3.7												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	7.5	10.1	13.9	19.7	23.9	26.9	29.4	28.9	26.1	21.2	15.3	9.4
	DBStd	8.34	4.02	5.07	4.57	3.92	2.69	2.53	1.90	1.90	2.97	3.18	3.88	3.51
	HDD10.0	231	100	57	14	1	0	0	0	0	0	0	6	53
	HDD18.3	1144	336	234	150	31	1	0	0	0	0	11	104	279
	CDD10.0	3665	23	61	137	291	431	506	603	587	482	347	166	33
	CDD18.3	1536	0	4	13	71	173	256	344	328	232	99	14	1
	CDH23.3	15374	3	37	101	471	1276	2483	4276	3796	2185	649	96	3
	CDH26.7	6579	0	3	16	129	429	1002	2134	1763	906	182	16	0
(14) Wind	WSAvg	1.5	1.5	1.6	1.5	1.5	1.4	1.5	1.7	1.6	1.6	1.6	1.5	1.5
(15) Precipitation	PrecAvg	1719	66	112	187	252	291	296	114	108	91	82	69	44
	PrecMax	2335	219	260	332	467	633	693	395	389	330	268	232	125
	PrecMin	1138	7	7	52	72	102	112	18	21	5	0	2	3
	PrecStd	323	52	55	78	99	123	149	76	71	71	70	55	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	26.8	28.5	32.0	34.2	35.6	37.9	37.5	36.1	32.8	28.7	22.9
		MCWB	16.5	18.6	20.2	22.7	23.7	26.0	26.1	26.1	25.7	23.4	20.4	16.4
	2%	DB	19.4	24.2	26.2	29.9	32.5	34.3	36.4	36.2	34.6	30.5	26.2	19.6
		MCWB	14.0	17.6	18.8	21.8	23.2	25.7	26.2	26.1	25.5	22.2	19.3	13.7
	5%	DB	16.7	21.2	23.7	28.1	30.8	33.1	35.4	35.0	33.2	28.7	23.8	17.2
		MCWB	12.1	16.1	18.0	21.1	23.1	25.5	26.2	26.2	25.2	20.8	17.6	12.7
	10%	DB	13.9	18.0	21.3	26.2	29.1	31.8	34.3	33.7	31.6	27.0	21.7	15.3
		MCWB	10.6	14.4	17.1	20.5	23.1	25.4	26.1	26.2	24.6	20.2	17.0	11.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	19.9	21.7	24.1	26.2	27.4	27.9	27.7	27.2	24.8	22.5	18.0
		MCDB	20.9	25.2	26.0	29.2	30.3	31.8	33.7	33.5	32.4	29.6	26.4	21.6
	2%	WB	15.1	18.2	20.3	23.1	25.4	26.9	27.3	27.2	26.5	23.7	20.8	15.8
		MCDB	18.5	23.0	24.4	27.8	29.3	31.5	33.1	32.8	31.7	27.8	23.5	17.8
	5%	WB	13.1	16.6	19.1	22.4	24.7	26.4	27.0	26.9	26.0	22.9	19.6	13.7
		MCDB	15.7	20.6	22.6	26.3	28.3	30.9	32.7	32.3	31.1	26.5	22.1	16.3
	10%	WB	11.0	14.8	17.7	21.6	24.1	26.0	26.6	26.5	25.5	22.0	18.0	12.1
		MCDB	13.6	17.7	20.4	24.9	27.1	30.2	32.0	31.6	30.0	25.2	20.9	14.4
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	7.5	7.9	7.9	7.1	8.6	8.4	8.3	8.9	8.7	8.4
		MCDBR	13.2	13.7	13.1	11.9	11.7	9.7	10.6	10.8	10.7	11.9	13.1	13.6
	5% WB	MCWBR	7.6	7.2	6.1	4.6	3.9	2.7	2.5	2.4	2.7	3.9	5.8	7.6
		MCWBR	10.7	11.9	10.2	9.4	8.1	8.0	9.4	9.4	8.3	9.2	9.2	9.6
(40) Clear-Sky Solar Irradiance	taub		0.647	0.694	0.779	0.751	0.651	0.617	0.577	0.625	0.635	0.657	0.605	0.592
		taud	1.590	1.543	1.441	1.530	1.753	1.839	1.951	1.810	1.762	1.654	1.712	1.699
	Ebn at Noon		589	606	587	626	693	713	742	703	679	624	620	610
		Edh at Noon	234	264	308	288	230	210	188	215	220	232	205	202
(44) All-Sky Solar Radiation	RadAvg		2.13	2.46	2.81	3.40	3.90	4.06	5.52	4.88	4.19	3.62	2.74	2.47
	RadStd		0.49	0.58	0.42	0.51	0.55	0.48	0.46	0.41	0.40	0.62	0.47	0.53

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.30	N/A	N/A	N/A	+0.20	+0.32	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.37	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	+140	+99

Nomenclature: See separate page